Enzo Harquin

PhD Student in Computer Science

Champs-sur-Marne, France eharquin

Summary

- PhD student at the Gaspard-Monge Computer Science Laboratory (LIGM), specializing in geometric algebra, computer vision, and computer graphics.

Experience

Gaspard-Monge Computer Science Laboratory (LIGM), PhD Student — Gustave Eiffel University, France
Research on geometric algebra and its applications to computer vision and computer graphics, with a particular focus on Projective Geometric Algebra.

Gaspard-Monge Computer Science Laboratory (LIGM), Research Intern — Champs-sur-Marne, France
Apr 2024 – May 2024 — 2 months
Parametric curves and surfaces in geometric algebras.
- Supervised by Vincent Nozick and Venceslas Biri.

Education

Gustave Eiffel University, Image Science — Champs-sur-Marne, France
Sept 2023 – 2024
- Graduated with honors.
- Ranked joint first among 19 students.

University of Technology of Compiègne (UTC), Computer Engineering, Software Engineering specialization — Compiègne, France
Sept 2020 – Sept 2023

Projects

Voxel Rendering Engine in Vulkan/C++ — Nov 2023 – Mar 2024
Research project for the Master's Degree in Image Science.
- Literature review, implementation, and comparative study of voxel rendering methods.
- Rasterization, ray tracing, and ray marching.

GAMUT
Interactive visual environment for geometric algebra constructions, research, exploration, and teaching.
- Expression-based construction and interactive manipulation of geometric objects.
- Support for multiple algebra models, including VGA, PGA, CGA, ACGA, and CCGA in two and three dimensions.
- Export of constructions as SVG and PNG images.

Skills

Research Areas

Programming Languages

Scientific Tools and Environments

Languages

French
Native speaker

English
C1 level — Cambridge English Linguaskill certification